

Quiz1

1-What relationship should Maintenance and Reliability Teams have with customers and suppliers for optimum effectiveness? Please choose the correct answer.

- A. Purchasing should be the only communicators with suppliers
- B. Management should be the only communicators with customers and suppliers
- C. Team members should be involved in communicating with customers and suppliers**
- D. Sales should be the only department communicating with customers.

2-the failure rate of a group of 10 identical electric motors is on failure every 400 hrs at operation of all motors in the system must operate

Calculate the availability where it takes 100 hrs for repair the motor

- A-0.75
- B-0.8**
- C-0.975
- D-1.2

3-What is the statistical average and the population standard deviation of the following series of numbers? 1.30, 1.25, 1.50, 1.12, 1.35, 1.38 Please choose the correct answer.

- A. Statistical average = 1.32; standard deviation= 0.128**
- B. Statistical average = 1.50; standard deviation= 0.25
- C. Statistical average = 1.32; standard deviation= 1.50
- D. Statistical average = 0.128; standard deviation= 1.32

4-What is the relationship between leading indicator metrics and lagging indicator metrics in the field of Maintenance and Reliability? Please choose the correct answer.

- A. Lagging indicators are measures that reflect accomplishment of goals, while leading indicators are measures of what must be done to improve results.
- B. Leading indicators are measures that reflect accomplishment of goals while lagging indicators are measures of what must be done to improve results.**
- C. Leading indicators are low level measures that reflect accomplishment of goals, while lagging indicators are measures of what must be done to improve results.
- D. Leading indicators are used by management, while lagging indicators are used by personnel on the shop floor.

5-As the maintenance manager at a new plant (facility) you are developing the maintenance workforce strategy for start-up. Select the best workforce attributes for maximum effectiveness, assuming that unit labor costs are equal? Please choose the correct answer.

- A. An annual bid and selection of a maintenance labor contractor is performed, using unitcosting of tasks normally required.
- B. A managing contractor is brought on site to handle all aspects and provision of maintenance labor.
- C. Multi skilled mechanical crafts (trades) and instrument electrical crafts (trades) are hired and developed.
- D. Operations performing minor maintenance and company multi-skilled technicians performing complex maintenance tasks**

6-The Weibull distribution with a shape factor less than 1 is associated with burn in or early life failures that could be caused by:

- A. Poor quality assurance.
- B. Irregular loading.
- C. Increasing strength.
- D. Improper assembly.

7-For which of the following elements is the maintenance manager directly responsible?

- A. Equipment failure probability
- B. Equipment operating practices
- C. Life cycle costs of equipment
- D. Optimization of maintenance costs

8-A maintenance manager has a spreadsheet list of all of the equipment failures that have occurred at the plant (facility) over the past three years. Which of the following analytical methods or tools might be used to sort and prioritize the list?

- A. Root Cause Failure Analysis
- B. Life Cycle Cost Analysis
- C. Total cost of ownership
- D. Pareto Analysis

9-Examples of workplace ethics issues include everything except:

- A. whistle-blowing.
- B. pay raises.
- C. conflicts of interest.
- D. workplace privacy.

10-What is one of the significant achievements of predictive maintenance programs?

- A. Permitting the ordering of long lead-time parts when the equipment breaks
- B. Minimizing use of components that are designed to wear out
- C. Increased awareness of and less uncertainty concerning machinery condition
- D. Providing delayed information about equipment condition after the equipment has failed

11-What is at the heart or core of the Organization for International Standards (ISO) 9000 series documents concerning quality of products and services?

- A. Management of quality assurance processes
- B. Human safety and environmental protection
- C. How to write quality control documents
- D. Procedures, document existence, compliance, and control

12-The most effective method of performance evaluation is:

- A. a field review process.
- B. a behaviorally anchored rating scale process
- C. a employee written assessment.
- D. a feedback process.

13-How do you avoid the pitfall of the Maintenance and Reliability Program appearing to have little value to the organization?

- A. Implement all available predictive technology techniques
- B. Use a systematic approach to ensure business goals drive the effort
- C. Establish a comprehensive communication program including e-mail web page, and newsletter
- D. Claim the success of the company is directly related to the Maintenance and Reliability Program

14-A Company produced 80.000 tons of "Good Product" last year, and the Sales department has projected a 5% increase in sales during the coming year. Production Statistics:

- 1.The production line operates 24 hours per day. seven days per week, except for the downtime listed below.
- 2.The production line downtime for maintenance averaged 108 hours per month during the past year.
- 3.The production line was shut down three times during the year due to power interruptions, for a total of 18 hours of downtime.
- 4.The production operators' union has threatened a strike that may last as long as three weeks if it occurs.
- 5.The production rate is expected to stay the same as last year's production rate.
- 6.Power interruption downtime is expected to stay the same as last year.

Based on the Production Statistics shown, what is the required production line availability to meet the sales projected for next year?

- A. 86.50%
- B. 88.50%
- C. 89.50%
- D. 93.50%

15-As the maintenance manager for a manufacturing company, you have conducted a study that reveals the mechanics, on average, spend four hours per ten hour shift on wrench time (tool turning time). Your first action should be:

- A. Improve planning and scheduling to increase wrench time (tool turning time).
- B. Encourage your planners to develop more aggressive job estimates.
- C. Inform management that four hours wrench time (tool turning time) per day is considered good.
- D. Investigate the barriers causing loss of wrench time (tool turning time).

16-When developing and presenting a new strategic plan to stakeholders, which of the following is the most important?

- A. Implementation timetable
- B. Gaining the trust of stakeholders
- C. Craft (trade) flexibility
- D. Establishment of a common goal

17-The technique which the shop floor supply the work when they get the work order request when needed

- A-push
- B-pull
- C-feedback
- D-management

18-Total quality management focuses all employees on producing products that meet customer needs. This is done by:

- A. Sharing information with all levels in the organization
- B. Developing a high level of expertise in all employees
- C. Eliminating processes that waste time and materials
- D. Balancing the needs of all stakeholders in the organization

19-Preventive maintenance tasks should be grouped by?

- A. Parts required, task interval and lockout -tag out requirement
- B. Skill required, parts required and equipment availability
- C. Skill required, task and interval
- D. Skill required, task interval and lockout -tag out requirement

20-Which of the following methods best identifies maintenance problems caused by maintenance or operating work processes?

- A. Root Cause Failure Analysis
- B. Bottleneck Analysis
- C. Weibull Analysis
- D. Pareto Analysis

Quiz2

1. When is the most appropriate time to use strategic planning?

- A "When reacting to day to day variation in business"
- B "At the beginning of each year"
- C "When a significant change is being made to the business"
- D "When you have to reduce costs quickly"

2. It is important that a maintenance and reliability improvement business plan:

- A "Be developed by the plant manager"
- B "Be entered in the CMMS"
- C "Outlines a payback of the investment"
- D "Be a formal document linked directly to business objectives"

3. Maintenance and operations performance agreements usually do not include?

- A "Repair response times"
- B "Availability targets"
- C "Agree staffing levels"
- D "Call-out procedures"

4. What is the primary use of operations and maintenance performance agreements?

- A "Know who to blame when problems occur"
- B "Know who is responsible for downtime"
- C "Deploy resources effectively"
- D "Cut down on maintenance errors"

5. An organization can achieve superior quality results by?

- "A: Continuous improvement of product or service"
- "B: Ensuring that quality controls are perfect"
- "C: Effectively marketing its products"
- "D:A&B"

6. $\text{Net Income} / \text{Total Assets} = ?$

- A "Net Profit"
- B "Return on Investment"
- C "Return on Net Assets"
- D "Return on Assets"

7. Gap analyses are based on?

- "Risk Analysis"
- "Employee surveys"
- "Quality Reviews"
- "Benchmarking"

8. Which of these is considered the best method of reducing noise?

- "Ear Plugs"
- "Blanket to block noise from room"
- "Box around noisy item to reduce noise output"
- "Noise transmission standards in equipment specifications"

9. A well-developed business case will include?

- "Health & Safety Impact"
- "Environmental Impact"
- "Customer Impact"
- "Financial Impact"

10. The plant manager desires to improve plant ROI. The maintenance manager should?

- "Wait for the plant manager to develop a plan"
- "Start deferring maintenance activities to save money"
- "Discuss how maintenance can help"
- "Develop and present a maintenance plan to increase ROI"

11. You can tell a safety program is working well because?

- "There are lots of good safety slogans"
- "There are lots of posters in the plant"
- "Periodic audits show adherence to policies & procedures"
- "Employees tell you everything is working well"

12. One of the most important steps after buy-in for creating an effective safety program is?

- "Have an extravagant kick-off event to get attention"
- "Examine and analyze work processes and working conditions"
- "Find a good safety manager"
- "Develop new policies and procedures for safety"

13. The best way to increase stakeholder involvement in asset maintenance is?

- "Increase the skill level of the operators"
- "Have maintenance staff do all maintenance"
- "Decrease the number of contractors in the facility"
- "Involve operations in the maintenance program"

14. Good work teams?

- "Have their mission and direction assigned by supervision"
- "Are assessed by supervision"
- "Are self-governing and responsible for performance"
- "All of the above"

15. Which is the most lagging indicator?

- "MTBF"
- "MTTR"
- "OEE"
- "Cost & Profitability"

16. What is the most important part of developing and presenting a new strategic plan?

- "Creating milestones and goals"
- "Getting craft to work on improvements"
- "Gaining trust from stakeholders"

"Creating a vision"

17. Which method of budgeting affords the greatest control over the maintenance budget?

"Targeted annual reductions in spending"

"Adding a percentage to historical spending"

"Budget controlled by operations"

"Activity-based costing"

18. The best way to show the value of Maintenance and Reliability Programs is to?

"Implement all available PdM technology"

"Systematically use business goals to drive the programs"

"Establish a comprehensive communication program"

"Claim company success is directly related to the program"

19-Shutdown Work Ratio is defined as the:

A. the amount of planned labor hours for key resource person(s) relative to actual labor hours worked during an outage.

B. amount of available work time key resource person(s) can work relative to the actual calendar time of the shutdown.

C. amount of work in manhours that a maintenance team actually achieved in a shutdown relative to the amount of work it could achieve.

D. the amount of estimated labor hours that a maintenance team can devote to an outage relative to the total downtime scheduled.

20-Which answer is not one of the steps in performing an RCA (Root Cause Analysis)?

A. Track the recommended solutions to ensure effectiveness.

B. Develop solutions and recommendations.

C. Identify possible mitigating factors.

D. Define the problem or the failure.

Quiz3

1. Which indicator is not a leading indicator?

"% planned maintenance"

"Asset Availability"

"Planner to craft workers ratio"

"% rework"

2. What are the mean and standard deviation of (1.3, 1.25, 1.50, 1.12, 1.35, 1.38)

"Mean = 1.32; SD = .128"

"Mean = 1.6; SD = .25"

"Mean = 1.32; SD = 1.5"

"Mean = .128; SD = 3"

3. The inherent reliability of a system is set during?

"Design"

"Installation"

"Operation"

"Maintenance"

4. The maintenance manager is directly responsible for

"Optimizing maintenance costs"

"Plant productivity"

"Life-Cycle Costs"

"Equipment failure prevention"

5. Maintenance carried out by the production department is known as?

"Total Productive Maintenance (TPM)"

"Operator Based Maintenance"

"Autonomous Maintenance"

"Preventive Maintenance"

6. Overall Equipment Effectiveness is a function of?

"Availability x Quality Rate x Throughput Rate"

"Availability x Reject Rate x Throughput Rate"

"Availability x Throughput"

"Availability x Quality Rate x Reject Rate"

7. Which of the following tasks does not help to incorporate changes to process & equipment

"Use an accepted risk analysis process"

"Use a database to record equipment failures"

"Use MOC policies and procedures to document change"

"Conduct safety, operability, and maintainability reviews"

8. Six sigma primarily uses the following manufacturing improvement tools?

"Root Cause Analysis"

"SPC & DOE"

"Failure Analysis"

"Pareto Analysis"

9. The supervision of a process designed to alter and improve the function of a business is?

"Management of Change"

"Change Management"

"Strategic Management"

"Management by Objectives"

10. The best way to increase stakeholder involvement in asset maintenance is?

"Increase the skill level of the operators"

"Have maintenance staff do all maintenance"

"Decrease the number of contractors in the facility"

"Involve operations in the maintenance program"

11. Which item is not important to identify customer requirements for M&R?

"Defining customer needs"

"Establishing standard operating procedures"

"Validating customer requirements"

"Identifying and prioritizing requirements"

12. Quality or "fitness for use or purpose" is the result of?

"Upper management direct involvement in quality issues"

"Good sales interface with clients"

"Strong quality control"

"The work of all departments and enforced by quality control"

13. Which metric measures the overall effectiveness of a safety program?

"First aid cases"

"Job Safety Analysis"

"Near Misses"

"Lost time injuries"

14. Which is the most lagging indicator?

"MTBF"

"MTTR"

"OEE"

"Operating Cost"

15. Which of the following is not a tool or technique to identify M&R improvements?

"TPM"

"RCM"

"Historical analysis of spending patterns"

"Process benchmarking"

16. The core of ISO 9000 documents for product and service quality is?

"Management of quality assurance processes"

"Human safety and environmental protection"

"Quality Control"

"Procedures, document existence, compliance, and control"

17-One of our five pumps fails each month. We have been getting 45 days of service out of our dryer belts. We have to fix five piping leaks per month. We have had to replace three of our 25 air cylinders in the last two years. If a Pareto diagram was created for this failure data, which equipment type would show the highest failure frequency?

- A. Air cylinders
- B. Dryer belts
- C. Pumps
- D. Piping

18-Quality superiority of a product or service can be maintained if the organization:

- A. Ensures its marketing and field service quality controls are perfect.
- B. Holds product specifications constant and historically accurate.
- C. Continually promotes its claims to superior quality and market leadership.
- D. Continues product or service improvement and has effective market promotion

19-Business excellence requires that in most companies the following things be done well and integrated effectively:

- A. budgeting, executing, follow-up
- B. communications, recruitment
- C. reliability improvement, sales, installation
- D. marketing, manufacturing, research and development

20-What should be done first to address a gap between your manufacturing performance and World Class performance?

- A. Develop gap closure plans and begin to implement them
- B. Assign additional resources to work on attaining World Class performance
- C. Define the value of the gap to determine its worth to the company
- D. Review gap analysis to ensure it reflects your current performance accurately

Quiz4

1. Which type of failures should be avoided by maintenance departments?

"Design Induced"

"Self-Induced"

"Machine Induced"

"Manufacturer Induced"

2. What is the most important thing to consider when deciding whether or not to stock a spare?

"Is the equipment on the criticality list?"

"Does the OEM have it on the suggested spares list?"

"Would there be significant loss of production?"

"Are we trying to reduce stores inventory?"

3. What is the order of the evolution of maintenance?

"RCM, PdM, PM, Reactive"

"PM, RCM, PdM, Breakdown"

"Breakdown, PdM, PM, RCM"

"Reactive, PM, PdM, RCM"

4. What are the most important activities on a PdM technicians' route?

"Collecting measurements in proper sequence"

"Recording measurements on all operating equipment"

"Observing and making notes on machine condition"

"Taking measurements in velocity, displacement, and acceleration."

5. Different materials appear to have different surface temperatures when they are the same when?

"They have different convective coefficients"

"They have different emissivities"

"They have different thermal capacities"

"They have different conductivities"

6. Maintenance tasks should be allocated to?

"Operations for minor tasks and maintenance for complex tasks"

"Operations or maintenance depending on availability"

"Multi-Skilled maintenance technicians"

"Contract maintenance personnel"

7. What is the best way to avoid premature leaks with new pump installations?

"Have experienced maintenance personnel from the design review till startup"

"Participate in the supplier's standards development process"

"Ensure the pump has a leak proof enclosure placed around it"

"Review the pump manufacturer's past performance record"

8. Which two factors have the greatest impact on the accuracy of radiometric imager readings?

"Relative Humidity & ambient air temperature"

"Emissivity and background temperature"

"Aperture and brightness"

"Range & span"

9. What time interval is most important for determining condition monitoring intervals?

"Time since last PM"

"Decay time from potential failure to functional failure"

"MTBF plus a safety margin"

"MTBF minus a safety margin"

10. Which analytical method is used to analyze equipment failure records?

"RCA"

"RCM"

"LCCA"

"Pareto Analysis"

11. Which of these is not a condition monitoring technique?

"IR inspection of a bus bar"

"Changing an air filter on mileage"

"Analyzing gearbox oil"

"Measuring tire tread depth"

12. Which of the following statements is true?

"Reliability can be improved by federal regulations"

"Designed in redundancy can improve reliability"

"Testing will improve reliability"

"No level of maintenance can increase inherent reliability"

13. Unit A is in series with Unit B. Unit A reliability = 0.40, Unit B reliability = 0.32.

"Overall Reliability = 0.128"

"Overall Reliability = 0.246"

"Overall Reliability = 0.365"

"Overall Reliability = 0.721"

14. Common reliability specifications on purchase orders are?

"Availability and Maximum Required Outage Time"

"Guaranteed Uptime and Guaranteed Downtime"

"MTBF & MTTR"

"MTBR and Guaranteed Availability"

15. Which factors are included in criticality analysis?

"Starts per year and cost to operate"

"Probability of failure and severity of consequence"

"Equipment Age and annual maintenance costs"

"Probability of environmental consequences and cost"

16. The interval between CBM tasks should be determined based on?

- "Time from potential failure to functional failure"
- "The failure rate under operating conditions"
- "Equipment criticality and impact of failure mode"
- "Failure rate under operating conditions"

17. What is the inverse of the MTBF

- "Failure Rate"
- "Failure Mode"
- "MTTF"
- "Service Life"

18-Preventive maintenance tasks should be grouped by?

- A. Parts required, task interval and lockout -tag out requirement
- B. Skill required, parts required and equipment availability
- C. Skill required, task and interval
- D. Skill required, task interval and lockout -tag out requirement

19-When would a strategic plan for Maintenance and Reliability be most appropriately required?

- A. Planning and scheduling are not being properly implemented at that plant (facility).
- B. Improved plant (facility) throughput is required to allow the business to commit to the requirements of a new customer.
- C. The markets that the company operates in are changing on a weekly basis.
- D. Fixed costs need to be reduced by 20% within the next year.

20-Which of the following is least likely to be linked to Maintenance and Reliability process improvements?

- A. Improved product design
- B. Reduced rework
- C. Lower operating costs
- D. Better availability capability

Quiz5

1. Which is not usually a primary motivator for workers?

- "Paycheck"
- "Desire for ownership"
- "Desire for competence"
- "Desire for recognition"

2. Which is the most effective method of performance evaluation?

- "A field review process"
- "A behaviorally anchored rating scale process"
- "An employee written statement"
- "A feedback process"

3. The four stages of performance encountered by new teams are; in order?

- "Warming/Absorbing/Performing"
- "Forming/Absorbing/Storming/Performing"
- "Forming/Storming/Norming/Performing"
- "Warming/Swarming/Absorbing/Norming"

4. Which are the best attributes of a strategy for creating a maintenance workforce

- "Annual bid by a maintenance contractor using unit costing"
- "Employing a managing contractor to handle all maintenance"
- "Employ/develop multi skilled mechanics & electricians"
- "Use operators and skilled craft depending on job complexity"

5. Which of the following best measures the effectiveness of the M&R workforce?

- "The size of the maintenance backlog"
- "A set of well-designed KPIs"
- "Trends in the M&R budget"
- "Man-hours of M&R work performed by production"

6. Which item is most overlooked when trying to implement an improvement program?

- "Assessing the impact of change on the work force"
- "Development of employee skills in root cause analysis"
- "Application of state-of-the-art CM technologies"
- "CMMS functionality"

7. What should a M&R professional do to ensure future success?

- "Accept standard operating and maintenance practices"
- "Attain expert knowledge of specific equipment and technology"
- "Stay abreast of current and emerging technologies"
- "Obtain multiple memberships in industry trade associations"

8. A study has revealed your organization has 40% wrench time. What should you do?

- "Improve planning and scheduling"
- "Encourage you planners to make more aggressive job estimates"
- "Inform management about how great your wrench time is"

"Investigate the barriers causing loss of wrench time"

9. Which is an appropriate KPI for a reliability engineer?

"Percent Rework"

"Equipment utilization rate"

"Shift stream factor and production rate"

"MTBF"

10. Failure to use protective eyewear on the job is an example of?

"Unsafe act"

"Unsafe condition"

"Willful safety violation"

"Unrecognized hazard"

11. What is the most likely cause of operations not being prepared for a scheduled down time?

"The maintenance manager did not follow up with the production MGR"

"The Production Manager didn't receive the email"

"The maintenance manager made a unilateral decision"

"The maintenance manager was unaware of the production schedule"

12. Reorganizations of the maintenance organization have yielded little change. You should first?

"Improve training materials"

"Develop and communicate a new strategic plan"

"Assess what gaps exist in processes and people"

"Establish KPIs"

13. Which should not occur during the performing stage of team development?

"Complete problem analysis"

"Begin to improve and control"

"Make changes to the process"

"Swing from optimism to uncertainty"

14. Who should be responsible for communicating and enforcing policy from management to the Shop Floor?

"First line supervisor"

"Middle Manager"

"Middle manager in the presence of the first line supervisor"

"Upper Management in the presence of the middle manager"

15. Examples of workplace ethics issues involve everything except?

"Whistle-blowing"

"Pay raises"

"Conflicts of interest"

"Workplace privacy"

16. Which has the least impact on gaining employee ownership of a maintenance task?

"Ask the employee to attend an M&R conference"

"Allow the employee to develop programs on critical assets"

"Ask the employee to develop programs on important assets"

"Assign work with details about how, when, and where to do it"

17-Which of the following is generally true?

- A) Construction contractors are usually equally skilled at performing all maintenance tasks.
- B) Maintenance is often reduced through project designs that provide in-place spares for all rotating equipment and heat exchangers.
- C) Reliability Centered Maintenance can be applied on capital projects in the pre construction stage to determine the maintenance plan.
- D) Most mechanics can easily alternate between doing capital project work and doing equipment diagnosis and repair work.

18-What is the best criterion for changing a known and controlled variable to meet a new customer requirement?

- A) Engineering analysis.
- B) Operator and maintainer experience.
- C) Senior management directive.
- D) Reliability and maintain ability analysis.

19-Weibull (life data) analysis is primarily used for?

- A) Maintainability analysis.
- B) Probability of failure determination.
- C) Life-cycle cost analysis.
- D) On time ration calculations.

20-The air supply to a pneumatic drill stamping center sometimes fails to meet requirements which lead to costly process interruptions. Four solutions have been proposed;

- A) Buy a larger new air compressor for \$40,000.
- B) Overhaul the existing compressor for \$10,000.
- C) Investigate the performance of the current air supply system.
- D) Slow down the process to meet compressor ability.

Quiz6

1. Preventive maintenance tasks should be grouped by?

- "Parts, task interval & lockout/tag out requirement"
- "Skill required, parts, and equipment availability"
- "Skill required, task and interval"
- "Skill, task interval & Lockout/tag out requirement"

2. In the broadest view, a maintenance planner

- "Places first priority on determining parts & tools"
- "Writes procedures & gather tools/parts for the job"
- "Provides information to schedule & execute of work"
- "Ensures good control of vendors"

3. Completed work orders should be turned in?

- "At the end of the day"
- "At the end of the shift"
- "After completing a group of tasks"
- "Upon completion of the work"

4. What should the reliability engineer do when outside contractors are hired?

- "Assure contractor personnel are very experienced"
- "Assure contractors have documented training programs"
- "Assure contractors have proper safety paperwork"
- "Assure contractors have been trained in work tasks"

5. The planning function should focus primarily on?

- "Accurate estimates"
- "Materials availability"
- "Equipment records"
- "Future work"

6. Desired min stock = 500 units, usage = 100 units/week and lead time = 2 wks. The reorder point=

- "350 units"
- "600 units"
- "700 units"
- "1200 units"

7. What happens to inventory acquisition costs and holding costs when inventory turns increases?

- "Acquisition cost increases & holding cost increases"
- "Holding cost & variable cost increases"
- "Acquisition cost decreases & holding cost increases"
- "Acquisition cost increases & holding cost decreases"

8. Which consideration has the most impact for scheduling a shutdown?

"Keeping cost within budget"

"Limiting jobs to only those that require shutdown time"

"Full utilization of manpower during the shutdown"

"Delaying shutdown until failure forces the facility down"

9. Which one of the following work order characteristics add the most value to the business?

"Correct charge codes"

"Proper authorization"

"Closed in a timely manner"

"Work completed is well documented"

10. Which of the following work order characteristics adds the most to the business?

"Accurate estimate of time and cost"

"Properly characterized expense vs. capital"

"Correct equipment number used"

"All parts ordered & received before work begins"

11. Maintenance planning makes money for the business because?

"It reduces finished goods inventory."

"It improves quality yield"

"Reduces maintenance time and increases production capacity"

"It ensures mechanics are busy all day"

12. What is the primary objective of an inventory control system?

"Provide accounting and management of inventory"

"Balance the cost of inventory with the risk of stock outage"

"Provide maintenance items when required & at right place"

"Reduce and sustain maintenance costs"

13. Based on maintenance man-hours, top performing companies have what ratio of maintenance tasks?

"50% PM/PdM, 80% P&S 1 wk in ahead, 5% emergency"

"25% PM/PdM, 60% P&S 1 wk ahead, 15% emergency"

"95% PM/PdM, 90% P&S 1 wk ahead, 5% emergency"

"30% PM/PdM, 40% P&S 1 wk ahead, 20% emergency"

14-If a machine is run for 500 hours and five failures are observed during this period, what is the Mean Time Between Failure?

A) 500 hours

B) 0.01 hours

C) 2500 hours

D) 100 hours

15-Which of the following most effectively makes up the members of a Manufacturing Team?

- A) Sales/ Customer/ Production/ Supplier/ Senior Management
- B) Maintenance/ Engineering/ Production/ human Resources
- C) Production/ Maintenance/ Supplier/ Engineering
- D) Senior Management/ Production/ Human Resources

16-What is the most important purpose of performing risk evaluations.

- A) To determine what level of response is cost justified.
- B) To clearly define activities required to contain the risks.
- C) To identify what events may have major impact on the business.
- D) To assure compliance with company policies.

17-Of the following, what is the best method for measuring employee skills and training?

- A) Maintain a training and qualification database.
- B) Ask employees to keep a training notebook.
- C) Keep records of all formal training courses taken.
- D) Have employees complete annual self-evaluations.

18-What relationship should Maintenance and Reliability Teams have with customers and suppliers for optimum effectiveness?

- A) Purchasing should be the only communicators with suppliers.
- B) Supervision should be the only communicators with customers and suppliers.
- C) Team members should be involved in communicating with customers and suppliers.
- D) Sales should be the only department communicating with customers.

19-When acquiring mission-critical, major machinery, it is usually appropriate to use a detailed equipment specification to assure the best equipment is selected and purchased. Which of the following is the most appropriate approach to developing an equipment specification for a piece of equipment?

- A. Arrange for potential equipment suppliers to provide their own Sales and Engineering representatives to visit the plant (facility) where the equipment is to be installed, as often as necessary, to prepare an equipment specification
- B. Review applicable company specifications, as well as supplier and other machinery specifications. Develop a custom specification that provides detailed construction, installation, performance, and other criteria as determined by mission requirements. Confirm that machinery offered by suppliers will meet the criteria in the custom specification
- C. Review applicable specifications from appropriate industry groups, Be the American Petroleum Institute (API). National Electrical Manufacturer's Association (NEMA) etc and select the specification that best meets the business goals.
- D. Send potential suppliers' copies of all of the applicable specifications your company has developed, with a cover letter explaining that equipment provided must meet applicable points within the specifications provided. Review supplier quotes and purchase the equipment from the supplier whose quote best meets business goals.

20-What should be done first to address a gap between your manufacturing performance and World Class performance?

- A. Develop gap closure plans and begin to implement them
- B. Assign additional resources to work on attaining World Class performance
- C. Define the value of the gap to determine its worth to the company
- D. Review gap analysis to ensure it reflects your current performance accurately

Quiz7

1. Which of the following statements is true?

- a) RCM does not bother about the consequence of a failure
- b) RCM does not try to achieve the inherent reliability of a system
- c) Time-based Preventive Maintenance is the cornerstone of the RCM output
- d) RCM does not believe that failure rate increases with the age of the equipment

2. Which of the following statements is true in case of RCM?

- a) Age of individual components can not be determined in a statistical manner
- b) RCM believes that run-to-failure strategy is often the best strategy
- c) RCM is not bothered about consequences of a failure
- d) RCM is not interested in achieving the inherent reliability of equipment

3. Waddington Effect states:

- a) The performance of an equipment improves upon regular overhauls
- b) The failure rate goes up if an equipment is regularly maintained by time-based preventive maintenance
- c) The failure rate goes down if an equipment is regularly maintained by time-based preventive maintenance
- d) The failure rate remains the same with regular time-based preventive maintenance

4. The most important piece of information in RCM is:

- a) Number of critical equipment
- b) Failure Modes
- c) Mean Time Between Failures (MTTF)
- d) Mean Time To Repair (MTTR)

5. Most failures in a plant are:

- a) Early
- b) Wear Out
- c) Random
- d) Constant

6) Which of the following is not a failure mode?

- a) Bearing seized
- b) Bearing problem
- c) Shaft sheared
- d) Circuit opened

7) Which of the following must not be included in the Function of a machine?

- a) Design Data
- b) What the user wants an equipment to do?
- c) Parameters that define the standard of performance
- d) Operating Context

8) Secondary Functions are:

- a) Additional functions an equipment is supposed to do

- b) Functions of other machinery that ensure performance of a critical machine
- c) Functions that are undesirable
- d) Functions that are desirable but not essential for the performance of a machine

9) Which strategy might be the cornerstone of an RCM strategy?

- a) Time-based maintenance (scheduled replacement/repair)
- b) Condition Monitoring or Condition Based Maintenance (CBM)
- c) Detective Maintenance (Inspections of Hidden Failures)
- d) Design Out Maintenance (DOM)

10) To improve the inherent reliability of a system the best strategy is:

- a) Condition Based Maintenance
- b) Design Out Maintenance
- c) Time-based Maintenance
- d) Run-to-Failure

11-When would a strategic plan for Maintenance and Reliability be most appropriately required?

- A. Planning and scheduling are not being properly implemented at that plant (facility).
- B. Improved plant (facility) throughput is required to allow the business to commit to the requirements of a new customer.
- C. The markets that the company operates in are changing on a weekly basis.
- D. Fixed costs need to be reduced by 20% within the next year.

12-Which of the following is least likely to be linked to Maintenance and Reliability process improvements?

- A. Improved product design
- B. Reduced rework
- C. Lower operating costs
- D. Better availability capability

13-When acquiring mission-critical, major machinery, it is usually appropriate to use a detailed equipment specification to assure the best equipment is selected and purchased. Which of the following is the most appropriate approach to developing an equipment specification for a piece of equipment?

- A. Arrange for potential equipment suppliers to provide their own Sales and Engineering representatives to visit the plant (facility) where the equipment is to be installed, as often as necessary, to prepare an equipment specification
- B. Review applicable company specifications, as well as supplier and other machinery specifications. Develop a custom specification that provides detailed construction, installation, performance, and other criteria as determined by mission requirements. Confirm that machinery offered by suppliers will meet the criteria in the custom specification
- C. Review applicable specifications from appropriate industry groups, such as the American Petroleum Institute (API), National Electrical Manufacturer's Association (NEMA) etc and select the specification that best meets the business goals.
- D. Send potential suppliers' copies of all of the applicable specifications your company has

developed, with a cover letter explaining that equipment provided must meet applicable points within the specifications provided. Review supplier quotes and purchase the equipment from the supplier whose quote best meets business goals.

